WORKED SOLUTION

STEP III 2012, Question 12

Probability and Statistics • Geometric Probability

STEP III 2012, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q12

The key idea

Splitting the triangle into three smaller triangles that share the vertex P shows that the three distances always add up to the height. For the minimum, the clean way in is the tail probability $\mathrm{P}(X > x)$. The event that every distance exceeds x is a smaller triangle similar to the original, and areas of similar figures scale as the square of the length ratio. In three dimensions the same idea works with volumes, which scale as the cube.

Solution

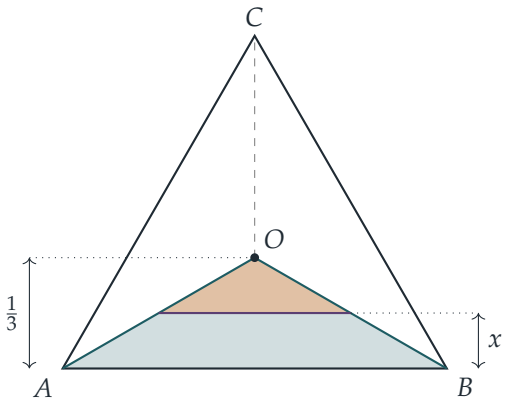
Part (i)

Let the side of the triangle be s. Since the height is 1, the area of ABC is $\frac{1}{2}s$. Joining P to the three vertices splits ABC into the triangles PAB, PBC and PCA. Their bases are the sides, each of length s, and their heights are x_1, x_2 and x_3. Adding the areas,

$$\tfrac{1}{2}sx_1 + \tfrac{1}{2}sx_2 + \tfrac{1}{2}sx_3 = \tfrac{1}{2}s, \qquad \text{so} \qquad x_1 + x_2 + x_3 = 1,$$

as required.

The region where X_1 is smallest. The points equidistant from AB and CA lie on the bisector of angle A. On the B side of this bisector, P is closer to AB than to CA. In the same way, P is closer to AB than to BC exactly when it lies on the A side of the bisector of angle B. In an equilateral triangle the angle bisectors are also the medians, and they meet at the centre O. So $X_1 = \min(X_1, X_2, X_3)$ exactly when P lies in the triangle ABO (shaded below). The distance from O to AB is a third of the height, $\frac{1}{3}$.



The density of X. The three triangles ABO, BCO and CAO are congruent, so each has probability $\frac{1}{3}$. (Ties between distances happen only on the bisectors, which have zero area.) By symmetry, X has the same distribution as X_1 conditioned on P lying in ABO. In that case $X = X_1$, the distance from P to AB.

? **Ask yourself** Inside triangle ABO, which points have $X_1 > x$?

They are the points above the line parallel to AB at distance x from it, which is the small amber triangle with apex O. It is similar to ABO, with height $\frac{1}{3} - x$ against $\frac{1}{3}$, so its area is a fraction $((\frac{1}{3} - x)/\frac{1}{3})^2 = (1 - 3x)^2$ of the area of ABO. Hence, for $0 \le x \le \frac{1}{3}$,

$$\mathrm{P}(X > x) = (1 - 3x)^2, \qquad \mathrm{F}(x) = \mathrm{P}(X \le x) = 1 - (1 - 3x)^2.$$

Also X can't exceed $\frac{1}{3}$, since the three distances add to 1. Differentiating,

$$\mathrm{f}(x) = \mathrm{F}'(x) = 6(1 - 3x) \quad (0 \le x \le \tfrac{1}{3}),$$

and $\mathrm{f}(x) = 0$ otherwise, as required.

Expected value.

$$\mathrm{E}(X) = \int_0^{1/3} 6x(1 - 3x)\,\mathrm{d}x = \Big[3x^2 - 6x^3\Big]_0^{1/3} = \tfrac{1}{3} - \tfrac{2}{9} = \tfrac{1}{9}.$$

Part (ii)

Let the four distances from the point to the faces be $Y_1, \ldots, Y_4$, and let $Y = \min(Y_1, \ldots, Y_4)$. Each face has the same area S, and the volume is $\frac{1}{3}S \cdot 1$. Joining the point to the four vertices splits the tetrahedron into four smaller tetrahedra. Each has a face as its base and one of the Y_i as its height, so

$$\tfrac{1}{3}S(Y_1 + Y_2 + Y_3 + Y_4) = \tfrac{1}{3}S, \qquad \text{so} \qquad Y_1 + Y_2 + Y_3 + Y_4 = 1.$$

The centroid is at distance $\frac{1}{4}$ from every face, so the inscribed sphere has radius $\frac{1}{4}$.

Ask yourself What does the set $\{Y > y\}$ look like now?

It is the set of points at distance more than y from all four faces. Each face moves inward by y, parallel to itself, which gives a smaller regular tetrahedron with the same centroid. The distance from the centroid to each face drops from $\frac{1}{4}$ to $\frac{1}{4} - y$, so the scale factor is $(\frac{1}{4} - y)/\frac{1}{4} = 1 - 4y$. Volumes scale as the cube of the scale factor, so for $0 \le y \le \frac{1}{4}$,

$$\mathrm{P}(Y > y) = (1 - 4y)^3.$$

(The same argument applied to the triangle in part (i) gives $(1 - 3x)^2$ directly: the inner triangle has inradius $\frac{1}{3} - x$.)

Since $Y \ge 0$, we can find $\mathrm{E}(Y)$ by integrating the tail probability:

$$\mathrm{E}(Y) = \int_0^{1/4} \mathrm{P}(Y > y)\,\mathrm{d}y = \int_0^{1/4} (1 - 4y)^3\,\mathrm{d}y = \left[-\tfrac{1}{16}(1 - 4y)^4\right]_0^{1/4} = \tfrac{1}{16}.$$

If you prefer to use the density, it is $12(1 - 4y)^2$ on $[0, \frac{1}{4}]$, and $\int_0^{1/4} 12y(1 - 4y)^2\,\mathrm{d}y = \frac{1}{16}$ as well.

So the expected distance to the nearest face is $\frac{1}{16}$.

Checks

- $\int_0^{1/3} 6(1 - 3x)\,\mathrm{d}x = 1$, so f is a genuine density.
- There's a pattern: in a triangle of height 1, $\mathrm{E}(X) = \frac{1}{9} = \frac{1}{3^2}$, and in a tetrahedron of height 1, $\mathrm{E}(Y) = \frac{1}{16} = \frac{1}{4^2}$. In one dimension, the distance from a random point of $[0, 1]$ to the nearer end has mean $\frac{1}{4} = \frac{1}{2^2}$, which fits.
- A Monte Carlo simulation with 4×10^5 random points in the triangle gave a mean of 0.1110 (against $\frac{1}{9} \approx 0.1111$), and a histogram that matches $6(1 - 3x)$. With 3×10^5 points in the tetrahedron, the mean was 0.0625 (against $\frac{1}{16} = 0.0625$).

Common slips

- Differentiating $\mathrm{P}(X > x)$ and forgetting the minus sign: the density is $-\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{P}(X > x)$.
- Forgetting that the inner triangle's height is $1 - 3x$, not $1 - x$: each of the three sides moves in by x.
- Using a squared scale factor in part (ii). Volumes scale as the cube.

Beyond STEP

Two tools from this question come up all the time in a first probability course: finding the distribution of a minimum through $\mathrm{P}(\min > x)$, and computing $\mathrm{E}(Y) = \int_0^\infty \mathrm{P}(Y >$

$y)\,\mathrm{d}y$ for a non-negative random variable. The pattern continues: in an n-dimensional simplex of height 1, the expected distance to the nearest face is $1/(n+1)^2$. Grimmett and Welsh, *Probability: An Introduction*, treats both tools carefully.

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